

Monte Carlo Denoising via Multi-scale Auxiliary Feature Fusion Guided Transformer

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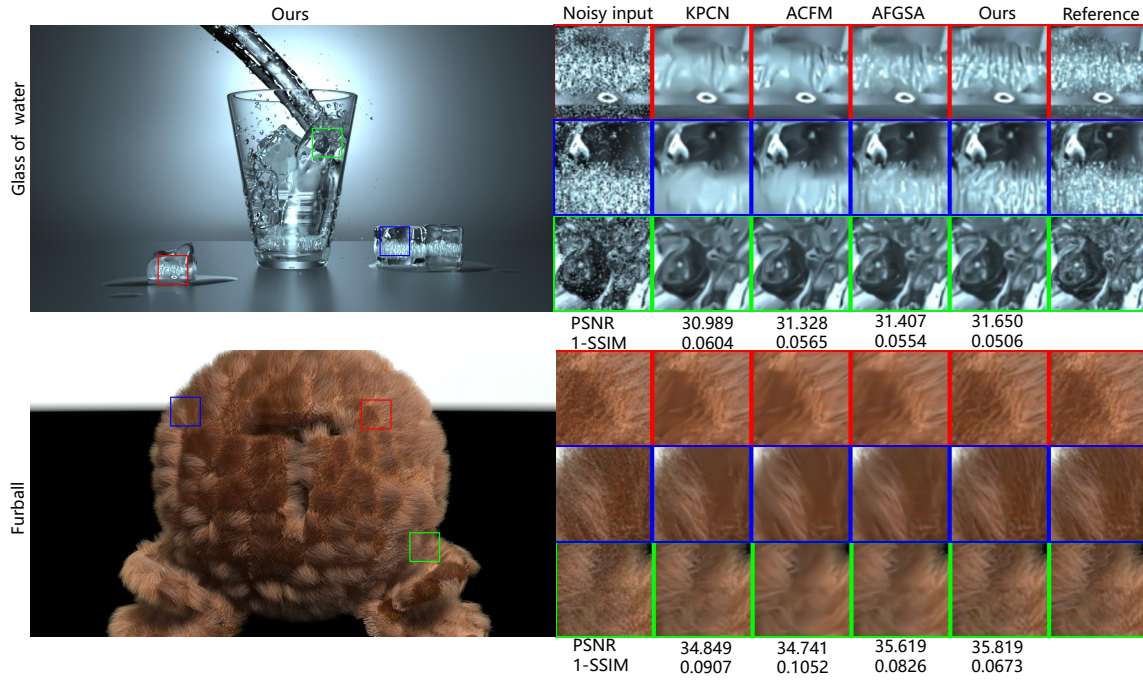


Figure 1: Employing a generative adversarial architecture, we introduce a transformer-based denoising network guided by multi-scale auxiliary feature. Our approach is compared with state-of-the-art methods, including KPCN [Bako et al. 2017], ACFM [Xu et al. 2019], and AFGSA [Yu et al. 2021]. The inputs, rendered with noise, are sampled at 128 samples per pixel (spp).

ABSTRACT

Deep learning-based single-frame Monte Carlo denoising techniques have demonstrated remarkable results in photo-realistic rendering research. However, the current state-of-the-art methods relying on self-attention mechanisms underutilize auxiliary

features and struggle to preserve intricate high-frequency details in complex scenes. Employing a generative adversarial architecture, we present a transformer-based denoising network guided by multi-scale auxiliary feature. The proposed U-shaped denoising network extracts multi-scale texture and geometric features from auxiliaries, modulating them to guide the improved transformer module's denoising process. The improved transformer module employs cross-channel self-attention to capture non-local relationships with near-linear computational complexity. Additionally, a gating mechanism is introduced in the transformer module's feed-forward network, enhancing information flow. Extensive experiments on noisy images with varied per-pixel sampling rates demonstrate the method's superiority in quantitative metrics and visual perception compared with state-of-the-art methods. Our method excels notably in intricate scenes with complex hair and texture details, which are historically challenging to denoise.

CCS CONCEPTS

• **Computing methodologies** → **Computer graphics**; *Rendering*; *Ray tracing*.

KEYWORDS

Monte Carlo denoising, Transformer, self-attention

ACM Reference Format:

Bingyi Chen, Zengyu Liu, Li Yuan, Zhitao Liu, Yi Li, Guan Wang, and Ning Xie. 2023. Monte Carlo Denoising via Multi-scale Auxiliary Feature Fusion Guided Transformer. In *SIGGRAPH Asia 2023 Technical Communications (SA Technical Communications '23)*, December 12–15, 2023, Sydney, NSW, Australia. ACM, New York, NY, USA, 4 pages. <https://doi.org/10.1145/3610543.3626179>

1 INTRODUCTION

Monte Carlo (MC) path tracing is a representative realistic image rendering technique widely used in various fields such as VFX, animation, and 3D games due to its generality and unbiasedness. Based on the MC estimation principle, when the samples per pixel (spp) is low, the sampled light distribution significantly deviates from the true light distribution, leading to a considerable amount of noise in the final image, while a high spp can produce extremely realistic images with a substantial computational cost. Therefore, denoising algorithms are commonly employed to post-process MC rendered images to produce noise-free results with negligible computational cost compared to the rendering stage.

The MC rendering process generates cost-effective supplementary data, referred to as auxiliary features, that encompass a rich collection of both geometric and texture information. Effectively extracting the valuable information from auxiliary features and leveraging it to reconstruct the noisy images constitutes a crucial challenge in MC denoising.

Recently, deep learning based MC denoising methods have gradually developed, among which notable contributions are the kernel-predicting convolutional network [Bako et al. 2017] and the adversarial MC denoising method [Xu et al. 2019]. Convolutional neural network models have achieved impressive denoising results but show limitations in capturing long-range pixel dependencies due to their limited receptive field.

Based on the aforementioned challenges, using a generative adversarial architecture, we present a transformer-based denoising network guided with multi-scale auxiliary feature for single-frame offline MC denoising tasks, aiming to leverage the capabilities of the transformer module to reconstruct more image details with multi-scale auxiliary features. Our denoising network employs a U-shaped architecture, incorporating Multi-scale Auxiliary Feature Fusion Guided Transformer (MSAFFT) modules and cascaded transformer modules at various levels to construct the encoder and decoder components, respectively. In order to fully leverage the abundant texture and geometry information of auxiliary features, we introduce a multi-scale auxiliary feature extraction module that can extract the rich hierarchical features in the auxiliary features and integrate them into the denoising network using the Conditioned Feature Modulation module (CFM) [Xu et al. 2019]. The transformer structure proposed performs self-attention across the channel dimension to capture long-range pixel relationships with near-linear computational complexity. Additionally, we introduce a gating mechanism in the Feed-Forward Network [Vaswani et al. 2017] to enhance the information flow within the network. In summary, our contributions and innovations are as follows: (1) We propose a multi-scale auxiliary feature extraction module based on residual dense blocks to effectively utilize the auxiliary information by extracting multi-scale features from the auxiliary features; (2) We propose a transformer module enhanced by multi-scale auxiliary feature fusion, incorporating self-attention across the channel dimension, and introducing a gating mechanism in the feed-forward network to enhance information flow throughout the network; (3) The proposed denoising method outperforms previous state-of-the-art methods in both visual perception and quantitative metrics, effectively removing MC noise, particularly for hair and regions with intricate details, leading to superior reconstruction results.

2 METHOD

We utilize a generative adversarial architecture and present a transformer-based denoising network guided by multi-scale auxiliary feature, and Fig. 2 shows the specific structure of the network. The objective of the generator in our network is to restore high-frequency details in noisy images and generate noise-free images that closely approximate the reference image; the discriminator aims to differentiate between the generated images produced by the generator and the reference images, and the discriminator in this paper adopts a multiscale PatchGAN structure [Liu et al. 2021].

2.1 Multi-scale Auxiliary Feature Extraction Module Based on Residual Dense Blocks

The features obtained after a single convolutional layer cannot fully utilize the original information of the image, resulting in the subsequent modules of the denoising network underperforming. We develop a multi-scale auxiliary feature extraction module, as shown in Fig. 2 (c). Specifically, after the downsampling operation, the auxiliary features are sent to the Auxiliary Feature Extraction Block (AFEB) of the corresponding level to obtain different scales of auxiliary features f_i . These multi-scale features are then incorporated into the denoising network to help reconstruct the noisy images. AFEB is composed of Residual Dense Blocks (RDB) [Wang

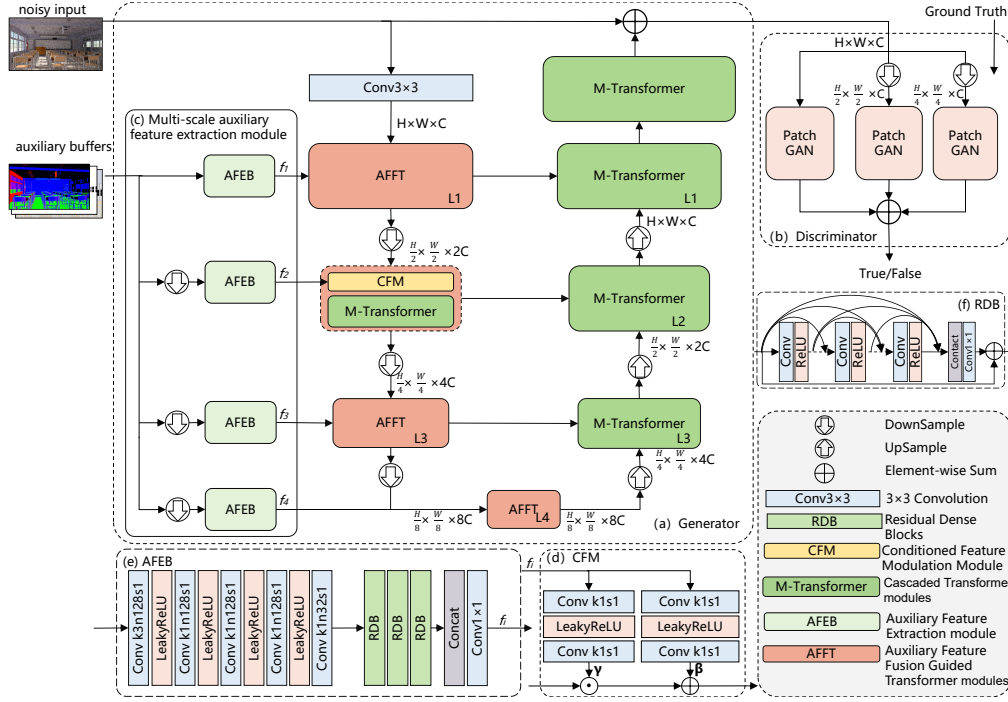


Figure 2: Network architecture. Our network structure is a generative adversarial network, aiming to effectively reconstruct the details of the noisy image under the guidance of the multi-scale auxiliary features. The noise input and auxiliary features are used as inputs to the denoising network and the multi-scale auxiliary feature extraction module, respectively. The multi-scale feature extraction module generates outputs that are fused with the denoising network through the CFM module, guiding the transformer module to improve image denoising.

et al. 2018]. As the depth of the AFEB network increases, the features obtained from each convolutional layer are gradually graded to obtain multi-level features. By combining both deep-level and shallow-level features, AFEB is able to extract valuable information from the auxiliary features at different scales more effectively.

2.2 Denoising Network

Pipeline. The denoising network needs to reconstruct the image under the guidance of these multi-scale features. As shown in Fig. 2 (a), the denoising network employs a U-shaped architecture to effectively capture both high and low frequency information and consists of a symmetrical multilevel encoder and decoder. The encoder integrates a multilevel MSAFFT module, while the decoder contains a multilevel cascaded Transformer module. The generator’s output is denoted as I' , which is obtained by adding the residual image R (output from the decoder) to the input noise image I : $I' : I' = I + R$.

Cross-channel self-attention. Self-attention is the main source of computational overhead in the Transformer module. The computational complexity of the traditional self-attention mechanism is $O(W^2 \times H^2)$ for a $W \times H$ pixel image. We implement self-attention across the channel dimension, which has nearly linear computational complexity. Concurrently, as shown in Fig. 3 (b), we introduce depth-wise convolution before computing the attention map to improve the ability to obtain local relationships. For the feature map

$M \in \mathbb{R}^{H \times W \times C}$ input by the Transformer module, we firstly apply a convolutional layer $W_{1 \times 1}$ with 1×1 convolutional kernel size to fuse the information across different channels, and the following deep-wise convolutional layer $W_{3 \times 3}$ not only enhance the network’s ability to capture local information, but also reduces the number of parameters in the network. This process generates the Q (Query), K (Key), and V (Value) matrices: $Q = W_{3 \times 3}^Q W_{1 \times 1}^Q M$, $K = W_{3 \times 3}^K W_{1 \times 1}^K M$, $V = W_{3 \times 3}^V W_{1 \times 1}^V M$. Then, by reshaping the Q and K matrices and convolving the two matrices, we obtain an attention graph of size $C \times C$ instead of a huge ordinary attention graph of size $(H \times W) \times (H \times W)$. $\hat{Q} \in \mathbb{R}^{HW \times C}$, $\hat{K} \in \mathbb{R}^{C \times HW}$, $\hat{V} \in \mathbb{R}^{C \times HW}$ is the dimensionally transformed Q , K , and V matrix, and α is the learnable parameter. The whole process is defined as equation 1.

$$SA(\hat{Q}, \hat{K}, \hat{V}) = \hat{V} \cdot \text{Softmax}\left(\hat{K} \cdot \frac{\hat{Q}}{\alpha}\right). \quad (1)$$

3 EXPERIMENTS

Implementation details. We utilized Wasserstein-GAN with a gradient penalty (WGAN-GP) to stabilize the training process. Our network was trained with Adam optimizer (batch size 8), and the learning rate was initialized at $2e-4$. It was halved after every 5k iterations up to 20k iterations. We ran our method using the PyTorch framework and trained the network on NVIDIA RTX 3090, which took about 6 days. We incorporated both Symmetric Mean Absolute

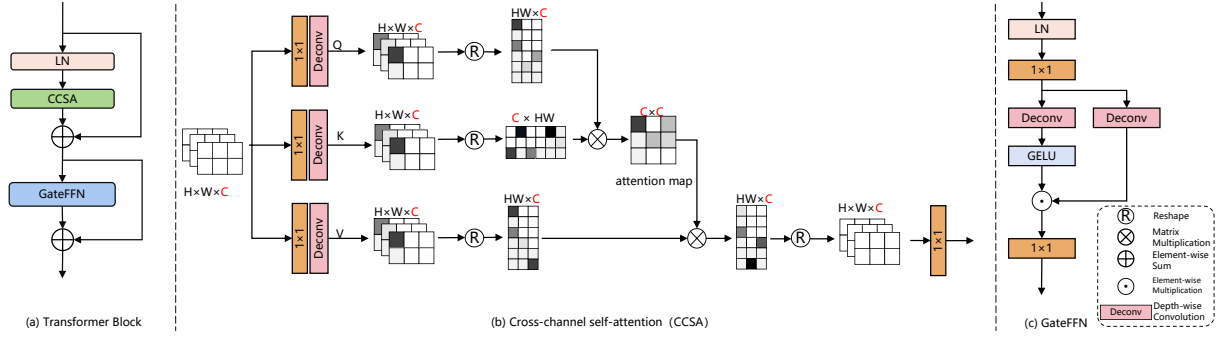


Figure 3: (a) Overview of the Transformer Block. (b) Cross-Channel Self-Attention module. (c) The feed-forward network with gating mechanism.

Table 1: Avg. denoising results (optimal results in bold).

Method	spp	PSNR $\uparrow$	1-SSIM $\downarrow$
KPCN [Bako et al. 2017]	8	32.596	0.0888
	16	34.577	0.0794
	32	36.219	0.0626
	64	37.689	0.0396
	128	38.967	0.0326
ACFM [Xu et al. 2019]	8	33.681	0.0767
	16	35.181	0.0669
	32	36.760	0.0534
	64	38.219	0.0422
	128	39.499	0.0332
AFGSA [Yu et al. 2021]	8	32.905	0.0797
	16	35.043	0.0648
	32	36.685	0.0519
	64	38.235	0.0421
	128	39.582	0.0346
MSAFFT(Ours)	8	33.816	0.0502
	16	35.809	0.0463
	32	37.508	0.0373
	64	39.097	0.0311
	128	40.452	0.0263

Percentage Error (SMAPE) loss and adversarial loss, with SMAPE effective in denoising low-frequency regions and adversarial loss focusing on high-frequency details of the image.

Dataset. We used the Tungsten renderer [Bitterli 2016] dataset during the experimental phase. It includes eight scenes with diverse materials and shapes, and each scene has approximately 200 sets of noisy and noise-free images along with auxiliary features.

4 RESULTS AND DISCUSSION

Leveraging a generative adversarial architecture, we present a transformer-based denoising network guided by a multi-scale auxiliary feature, and it is benchmarked against state-of-the-art techniques, including KPCN [Bako et al. 2017], ACFM [Xu et al. 2019],

and AFGSA [Yu et al. 2021]. These methods have publicly available code and model weights, we use the publicly available model weights to compare them on the same test set with the same experimental setup.

Table 1 shows the average test results for all scenes, where our approach outperforms previous methods numerically and produces better visual results. As illustrated in Fig. 1, our method is particularly effective in denoising scenes containing complex information such as hair and reflection details. In the hair and the Glass of water scene with complex transmission details, previous methods tend to cause blurring and loss of details. However, our presented method can better preserve these high-frequency details thanks to the design of the multi-scale network and the multi-scale auxiliary feature fusion.

ACKNOWLEDGMENTS

This work was supported by the Chengdu Science and Technology Project (2019-YF08-00285-GX), the National Natural Science Foundation of China (No. 61976156), and the Sichuan Science and Technology Program (2022NSFSC0640).

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